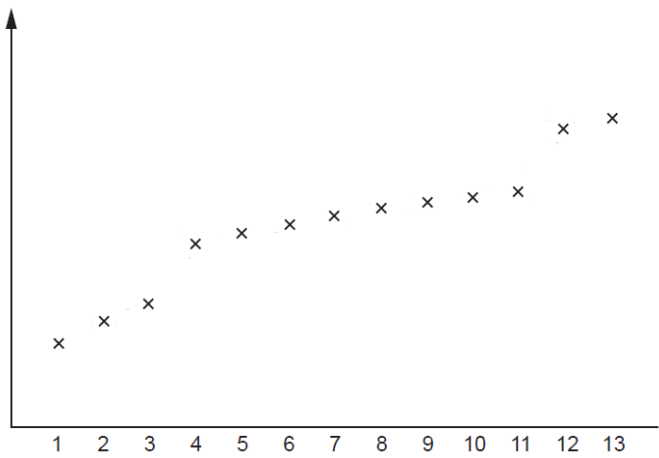


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8.	(a)			transfer of electrons / correct electronic structure of ions (1) charges on ions (1) Al $\times$ $\times$ $\times$ $\ddot{\text{O}}:$ $\ddot{\text{O}}:$ $\ddot{\text{O}}:$ $\longrightarrow$ $[\text{Al}]^{3+}$ $[\text{Al}]^{3+}$ $[\text{x}\ddot{\text{O}}:]^{2-}$ $[\text{x}\ddot{\text{O}}:]^{2-}$ $[\text{x}\ddot{\text{O}}:]^{2-}$		2		2		
	(b)			$n = \frac{pV}{RT}$ (1) award (1) for correct units for all of volume, temperature and pressure $V = 6.08 \times 10^{-5} \text{ m}^3$ $T = 493 \text{ K}$ $p = 1.01 \times 10^5 \text{ Pa}$ $n = \frac{1.01 \times 10^5 \times 6.08 \times 10^{-5}}{8.31 \times 493}$ $n = 1.499 \times 10^{-3} \text{ mol}$ (1) $M_r = \frac{0.400}{0.001499} = 266.8 \quad \Rightarrow \quad M_r \text{ of Al}_2\text{Cl}_6 \text{ is } 267$ (1)	1	2	1	4	3	

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)				1		1		
		(ii)	I	award (1) for any of following • general increase because there is a greater effective nuclear charge as more electrons are removed • general increase because the same number of protons are holding fewer and fewer electrons • general increase because electrons are being removed from an increasingly positive ion neutral answers – less shielding / electrons are closer to the nucleus	1			1		
			II	award (1) for either of following • sharp rises due to electron being removed from an inner shell / a shell which is closer to nucleus • sharp rises due to electron being removed from a shell which is closer to nucleus so has less shielding do not accept reference to sub-shells	1			1		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(d)	(i)		^{26}Mg		1		1		
		(ii)		0.0005 g (2) if answer not correct award (1) for either of following 8.0 mg = 0.008 g 2.88×10^6 is equivalent to 4 half-lives		2		2	2	
				Question 8 total	3	8	1	12	5	0